Optimal order placement in limit order markets

A technical summary and critical review of
Cont and Kukanov [2017]

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Abstract

In this report, we seek to provide a technical summary and critical review of *Optimal order placement in limit order markets* by Rama Cont and Arseniy Kukanov (2017). The paper addresses the problem of optimally splitting a trade slice between market orders and limit orders across multiple exchanges, in order to minimize expected execution cost while controlling execution risk. We detail the model's formal structure, the derivation of its cost function, and the main analytical and numerical results. We then place this contribution within the broader literature on optimal execution and market microstructure, and discuss its limitations as well as relevant extensions, that for some, have been proposed in subsequent work. As an extra, we provide a Jupyter notebook in which we implemented the article's algorithm and investigated some stylized facts at the following git repo : git-Algo-Trading.

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1. Introduction and motivation

In modern electronic equity markets, the execution of a large institutional order is rarely a single, atomic decision. It is instead decomposed into two nested problems: an *order scheduling* problem, which determines how to execute a parent order across time (to minimize market impact, *inter alia*), and an *order placement* problem, which determines, for each scheduled slice, how to best route that quantity across available venues and order types (namely market and limit). While the former has been extensively studied since the original work of Almgren and Chriss [2001], the latter has received comparatively less attention, despite being of direct practical relevance to trading actors such as brokers.

The order placement problem is fundamentally and ultimately a trade-off between *cost* and *certainty*. A market order executes immediately and with certainty, but consumes the spread and pays taker fees. A limit order posted at the best bid can earn the spread and receive a maker rebate, but faces the risk of not being filled within the execution horizon – forcing a potentially costly catch-up purchase at time T using a market order. This tension is further complicated in fragmented markets. In the US equity market, for instance, at least 11 active exchanges simultaneously displayed quotes for the same security in 2025, each with its own queue length, order flow statistics, and fee/rebate structure (even though NYSE and NASDAQ were the major ones, cf. Investopedia [2025]). For instance, a trader wishing to buy S shares over a one-minute horizon must simultaneously decide how much to meet immediately via market orders, and how to distribute passive limit orders across venues to maximize the probability of fill while minimizing trading costs and other risks that will be detailed in Section 2.1.

Cont and Kukanov [2017] is, to the best of our knowledge, the first paper to provide a rigorous and tractable mathematical framework for this multi-venue order placement problem. Positioned at the intersection of *optimal execution theory* and *market microstructure*, it complements the scheduling literature by formalizing the routing decision that follows once a schedule is fixed. It also extends the microstructure literature on limit order markets, which had previously focused on stylized single-unit, single-venue models, to a setting that is operationally closer to practice. The paper formulates order placement as a convex optimization problem, derives closed-form solutions for the single-exchange case, characterizes optima in the multi-exchange case through shortfall probabilities, and proposes a stochastic approximation algorithm for the general numerical case.

2. Model and framework

2.1. Setup, assumptions, and risk characterization

The model considers a trader who must buy $S > 0$ shares of a stock within a short time horizon $[0, T]$ (note that it could also be a selling problem). The horizon T is fixed and the quantity S represents a scheduled trade *slice* rather than the full parent order. At time 0, the trader may submit two types of orders: a market order of size $M \geq 0$ (routed to the cheapest available

venue), and limit orders $(L_1, \ldots, L_K) \geq 0$ posted simultaneously on $K \in \mathbb{N}^*$ exchanges at the National Best Bid. The full decision vector is:

$$X = (M, L_1, \ldots, L_K) \in \mathbb{R}_+^{K+1}$$

If, at time T, the total executed quantity falls short of S, the trader submits a catch-up market order for the residual so that the execution target S is always ultimately met.

Limit order execution mechanics. On each exchange $k \in [\![1, K]\!]$, there are already $Q_k \geq 0$ shares queued ahead of the trader at the best bid. Over $[0, T]$, the total order outflow from the front of that queue – comprising both incoming sell market orders and cancellations ahead – is modeled as a random variable ξ_k. Under standard FIFO (First In, First Out) priority, the fill on exchange k is:

$$F_k(L_k, \xi_k) = \min\left(\underbrace{(\xi_k - Q_k)^+}_{\text{net outflow}}, L_k\right) = (\xi_k - Q_k)^+ - (\xi_k - Q_k - L_k)^+$$

The fill is therefore strictly zero if the outflow fails to exhaust the queue ahead ($\xi_k \leq Q_k$), and equals L_k only if the outflow exceeds both the prior queue and the trader's own order size. Partial fills arise in the intermediate region $Q_k < \xi_k < Q_k + L_k$.

The total executed quantity at $t < T$ is then:

$$A(X, \xi) = M + \sum_{k=1}^{K} F_k(L_k, \xi_k)$$

Risk description. Several sources of cost and risk arise naturally in this framework:

- *Explicit trading costs*: market order taker fees f and limit order maker rebates r_k, in addition to the spread that is paid by the former and earned by the latter.
- *Adverse selection on limit order fills*: filled limit orders tend to be preceded by adverse price moves. The paper incorporates this via *effective rebates* $r_k = r_k^e + \mathrm{AS}_k$, where r_k^e is the posted exchange rebate and $\mathrm{AS}_k \leq 0$ is a venue-specific adverse selection penalty, empirically estimated as the average short-term price move following a fill at venue k.
- *Execution shortfall risk*: if $A(X, \xi) < S$, the trader must catch up with a market order at time T, at a cost that is typically higher than at time 0. A symmetric penalty applies if the trader overshoots the target and thus needs to liquidate the surplus position at $t = T$.
- *Market impact*: all submitted orders – whether filled or not – contribute to price impact, modeled linearly in the total submitted quantity.

2.2. Cost function and optimization problem

Building on the risk description of the previous section, the authors define the *scenario cost* $v(X, \xi)$ as the sum of all execution costs in a given realization ξ of queue outflows:

$$\begin{aligned} v(X,\xi) := \quad & \underbrace{(h+f)M}_{\text{market order cost}} \quad - \underbrace{\sum_{k=1}^{K}(h+r_k)F_k(L_k,\xi_k)}_{\text{limit order savings}} \\ & + \underbrace{\theta\left(M + \sum_{k=1}^{K} L_k + (S - A(X,\xi))^+\right)}_{\text{market impact}} \\ & + \underbrace{\lambda_u(S - A(X,\xi))^+}_{\text{underfill penalty}} + \underbrace{\lambda_o(A(X,\xi) - S)^+}_{\text{overfill penalty}} \end{aligned} \tag{1}$$

where h is the half bid-ask spread, $\theta > 0$ is the linear impact coefficient, and $\lambda_u, \lambda_o \geq 0$ are the marginal penalties (in dollars per share) for falling behind or exceeding the target S.

The paper operates under three mild parametric assumptions (A1-A3): that limit order fills are always beneficial net of adverse selection ($h + r_k > 0$), that overshooting the target is undesirable ($\lambda_o > h + \max_k r_k$), and that catching up at time T is more costly than using a market order at time 0 ($\lambda_u > h + f$). These ensure the optimization is well-posed and economically meaningful.

Several structural features of this objective deserve emphasis. First, the impact term is paid on *all submitted quantities* – including the catch-up market order at time T – which penalizes aggressive overbooking. Second, the penalties λ_u and λ_o don't need to be symmetric: a trader with time-sensitive information would set $\lambda_u \gg \lambda_o$ to reflect a high urgency. Third, the cost function is a sum of convex functions of X for any fixed ξ.

Then, the optimal order placement problem is naturally defined as:

$$\min_{X \in \mathbb{R}_+^{K+1}} V(X), \quad \text{where} \quad V(X) = \mathbb{E}[v(X,\xi)]$$

Under assumptions A1-A3, the authors prove (Propositions 1 and 2) that V is convex, bounded below, and has a global minimizer $X^* \in C$, where the feasible set is:

$$C = \left\{ X \in \mathbb{R}_+^{K+1} \;\middle|\; 0 \leq M \leq S,\ 0 \leq L_k \leq S - M,\ M + \sum_{k=1}^{K} L_k \geq S \right\}$$

Those constraints are notable: while the market order may not exceed S and each individual limit order may not exceed the remaining quantity $S - M$; the *aggregate* submitted quantity can – and optimally often does – exceed the target. This *overbooking* behavior reflects the fact that, with multiple uncertain limit orders, submitting more than S in aggregate is a rational strategy to hedge execution risk via diversification (cf. Section 3).

3. Main results

3.1. The single-exchange case

When $K = 1$, Proposition 1 implies $M^* + L^* = S$, so the optimization reduces to a one-dimensional problem over $M \in [0, S]$. Because there is no cross-venue diversification, overbooking is never optimal here. The problem is then differentiable and strictly convex, and Cont and Kukanov derive the following closed-form solution (Proposition 3):

Define the quantile $F^{-1}(p) := \inf\{x : P(\xi \leq x) \geq p\}$ of the queue outflow distribution. Then:

1. **Pure limit order regime:** if $\lambda_u \leq \dfrac{2h + f + r}{F(Q + S)} - (h + r + \theta)$, then $(M^*, L^*) = (0, S)$.
2. **Pure market order regime:** if $\lambda_u \geq \dfrac{2h + f + r}{F(Q)} - (h + r + \theta)$, then $(M^*, L^*) = (S, 0)$.
3. **Mixed regime:** otherwise, the optimal allocation is:

$$L^* = F^{-1}\left(\frac{2h + f + r}{\lambda_u + h + r + \theta}\right) - Q, \qquad M^* = S - L^*$$

This result is obtained by setting the derivative $V'(M) = 0$ and exploiting the monotonicity of V', which follows from the convexity of V. The derivative takes the explicit form:

$$V'(M) = 2h + f + r - (\lambda_u + h + r + \theta)\, F(Q + S - M)$$

and its zero corresponds precisely to the result derived above.

The economic interpretation is the following: replacing one passive share with one market share incurs an immediate certain cost of $h + f$, but eliminates the risk that the passive share fails to fill and triggers a catch-up costing $\lambda_u + \theta$, while forfeiting the conditional rebate gain $h + r$. The optimal limit order size is therefore determined by a *quantile* of the outflow distribution, not its mean. A limit order fills only when outflow is sufficiently large, so the tail of F governs here.

Another structural stylized fact is that L^* is bounded independently of S: as S grows, L^* saturates at a level determined by queue and outflow statistics, while excess quantity flows into market orders. Thus, the market order fraction M^*/S increases monotonically in S, reflecting the bounded passive execution capacity of any single venue.

3.2. The multi-exchange case

For $K > 1$, a full closed form is generally unavailable. Instead, Cont and Kukanov characterize the optimal allocation through first-order conditions (Proposition 4), which take the form of *execution shortfall* probabilities. Under proper continuity and interiority conditions, X^* is optimal if and only if:

$$P(A(X^*, \xi) < S) = \frac{h + f + \lambda_o + \theta}{\lambda_u + \lambda_o + \theta} \tag{2}$$

$$P\big(A(X^*, \xi) < S \mid \xi_j > Q_j + L_j^*\big) = \frac{\theta / P(\xi_j > Q_j + L_j^*) + \lambda_o - (h + r_j)}{\lambda_u + \lambda_o + \theta}, \quad j = 1, \ldots, K \tag{3}$$

These conditions are necessary and sufficient for optimality, thanks to the convexity of V. They are derived by computing the KKT conditions of the optimization problem and showing that, under the stated assumptions, all multipliers are zero at the interior optimum.

The interpretation of (2) and (3) is relevant. Equation (2) pins down the optimal *aggregate* shortfall probability: it must equal the ratio of the marginal cost of a market order to the total penalty range. Equation (3) then governs each venue's limit order size: the conditional shortfall probability, given that venue j's order filled completely, must reflect a venue-specific balance between the rebate gained and the impact incurred. Venues with higher effective rebates r_j, shorter queues Q_j, or heavier outflow tails result in larger (limit order) allocations.

3.3. Numerical algorithm and simulation results

For the general $K > 1$ case, the authors propose a *stochastic gradient descent* algorithm based on the robust stochastic approximation framework of Nemirovski et al. [2009]. The core of the method is that the gradient of $v(X, \xi)$ with respect to X involves only indicator functions of fill events and shortfall events – quantities that can be directly read off from past order execution outcomes without parametric assumptions on F and its distribution. This yields a non-parametric algorithm that updates order allocations iteratively as execution data accumulates:

$$X_n = X_{n-1} - \gamma\, g(X_{n-1}, \xi^n), \qquad \hat{X}_N^* = \frac{1}{N} \sum_{n=1}^{N} X_n$$

where $g(X, \xi) = \nabla_X v(X, \xi)$ and γ is a step size calibrated to problem parameters. The algorithm has a theoretical convergence rate of $O(1/\sqrt{N})$ and is shown numerically to converge within 1,000-10,000 iterations from arbitrary starting points. On a standard laptop, the authors argue that an allocation across $K = 12$ exchanges can be computed in under 200 milliseconds (see the notebook for a succinct implementation test, Section 4.4).

Some stylized facts seem worth noting (cf. Section 5.1 of the paper) as they yield several intuitive findings. The optimal market order fraction increases with target size S (confirming bounded limit order capacity), decreases with the number of exchanges K (as passive diversification opportunities expand), and increases with urgency λ_u. The overbooking tendency – total submitted quantity exceeding S – becomes more pronounced with more venues and less correlated order flows. For large S, the allocation becomes insensitive to rebates, as execution risk concerns dominate cost differences across venues.

Finally, Cont and Kukanov apply their algorithm to real world tick data. They calibrate the algorithm to MSFT (Microsoft) order flow on NASDAQ and BATS Z (another exchange) during Q1 2012 and evaluate it out-of-sample on April 2012 data. The optimal strategy consistently outperforms a naive equal-split benchmark by several cents per share, with performance gaps widening as execution risk aversion λ_u decreases (i.e., when there is more room to exploit passive execution opportunities).

4. Critical discussion

4.1. Placement in the literature

The contribution of Cont and Kukanov [2017] lies at the junction of two streams of research that had previously evolved largely in parallel. The *optimal execution* literature – following Bertsimas and Lo [1998] and Almgren and Chriss [2001] – is primarily concerned with the scheduling of orders over time. In these models, execution is typically represented through price impact functions and trading rates, while the detailed mechanics of order placement within the limit order book are abstracted away. In particular, these frameworks do not explicitly model the choice between market and limit orders or the queue dynamics that determine limit order execution.

On the other hand, the *market microstructure* literature (Foucault [1999], Parlour [1998], Rosu [2009]) has studied the limit/market choice in great theoretical depth, but under strong assumptions: traders submit single-unit orders, queues are absent or exogenously given, and the focus is on equilibrium price formation rather than execution optimization. Cont and Kukanov bridge these two traditions by constructing a model that is analytically tractable and implementable.

The paper also connects to the *smart order routing* literature. Moallemi et al. [2012] study routing of infinitesimal limit orders to a single destination in a reduced-form equilibrium. Almgren and Harts [2008] propose a market-order routing algorithm in the presence of hidden liquidity, while Laruelle et al. [2011] apply stochastic approximation to optimize across dark pools. Cont and Kukanov seem to be the first to combine these elements: finite order sizes, FIFO queue modeling, limit/market mix, multiple venues, and maker-taker fee structures, in a single unified framework.

4.2. Limitations

Despite its contributions, the model of course rests on several assumptions to allow tractability, that limit its applicability in practice. In particular, I have identified few different limitations that are worth noting.

Static order placement. The optimization is solved once at time $t = 0$, with a single catch-up market order at time $t = T$. In practice, traders continuously monitor queue positions, cancel and resubmit limit orders as conditions evolve. The authors acknowledge this, advocating that the static problem can be interpreted as the ergodic (average) version of a dynamic, stochastic control one. But the gap to a fully adaptive policy can be significant.

Single price level for limit orders. All limit orders are assumed to be posted at the National Best Bid. The model therefore does not consider the possibility of placing orders inside the spread – where execution probability is lower but potential gains may be higher – nor deeper in the order book. As a result, it does not capture strategies that diversify order placement across different price levels. No slippage event is considered either.

Exogenous and stationary queue outflows. The random outflows ξ_k are treated as exogenous, identically distributed across repetitions, and independent of the trader's own actions. In reality, queue dynamics are endogenous: large passive orders visibly contribute to the queue and may attract or repel order flow. Furthermore, outflow distributions are non-stationary, exhibiting strong intraday patterns and regime-switching behavior which may undermine the incremental learning logic of the algorithm.

Linear and symmetric impact. Market impact is modeled as linear in total submitted size and identical across order types and venues, while empirical studies show that impact is typically concave in order size and differs between market and limit orders (cf. Almgren et al. [2005]).

Calibration of λ_u, λ_o. The shortfall and overfill penalties strongly influence the optimal allocation. While the paper provides a theoretical anchor – they are the dual variables of expected shortfall and overfill constraints respectively – it offers no empirical procedure to estimate them from data. The tick-data application simply consider three *ad hoc* values without a calibration.

4.3. Extensions and related work

The convex, model-based approach proposed by Cont and Kukanov has since served as a baseline for some research that takes a rather different perspective. Rather than solving an explicit optimization problem under parametric assumptions on queue outflows, this literature explores whether an agent can *learn* an effective order placement policy directly from data, without specifying the distribution of ξ or calibrating the penalty parameters λ_u and λ_o, reflecting the broader shift toward data-driven methods in recent years.

An example of extension in this direction is conducted by Schnaubelt [2022], who studies limit order placement on cryptocurrency markets using deep reinforcement learning. The paper explicitly positions itself as a model-free alternative to the Cont and Kukanov [2017] framework, arguing that even if a fully realistic model of limit order markets could be specified, solving it analytically would be challenging in practice. Training on high-frequency data, the proposed agent (trained by proximal policy optimization) is shown to outperform several static strategies, even though the multi-exchange case seems not to be relevant here and is thus not studied.

More broadly, Hambly et al. [2023] review the growing use of reinforcement learning (RL) in financial decision problems through a survey paper. They emphasize that traditional stochastic control/analytical approaches rely on explicit modeling assumptions, whereas RL methods leverage large datasets with fewer structural constraints. Cont and Kukanov [2017] is, again, cited as

a model-based benchmark. The trade-off between the two approaches remains unsolved: model-based methods offer interpretability and analytical stylized facts, while data-driven methods provide flexibility at the cost of higher data requirements and potential robustness/overfitting issues.

4.4. Personal implementation

In addition to this report, we implemented the Cont and Kukanov framework and experimented with its numerical optimization procedure on simulated data. As this implementation was not explicitly required for the report, we do not discuss it in detail here. The corresponding notebook is nonetheless available as supplementary material and may be of independent interest.[1]

[1] https://github.com/malo-david/Algo-Trading-project-Cont-Kukanov-/tree/main

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